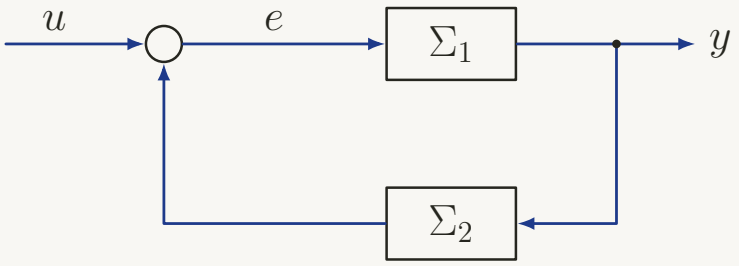
EXERCISE 04

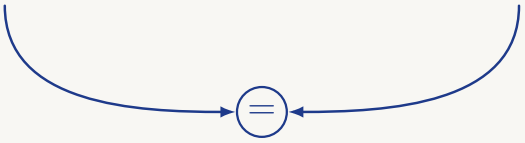
Time Response

1 | Block diagrams (PS01, Ex 1)



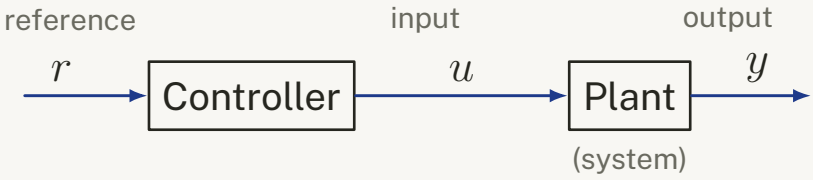
$y = \Sigma_1 u$ $\qquad$ $e = u - \Sigma_2 y$

$\rightarrow\ y = (I + \Sigma_1\Sigma_2)^{-1}\Sigma_1 u$ $\qquad$ $\rightarrow\ y = S_1(I + S_1S_2)^{-1}u$

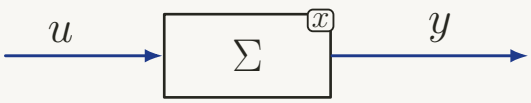


RECAP

General structure of our problem:

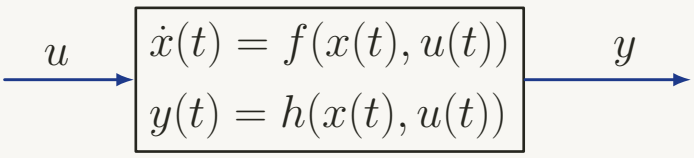


Focus on our general system:



Which is really just...

mathematical formulation
of our system



↪ we can find $f(x(t), u(t))$ & $h(x(t), u(t))$ through modeling.

BUT... In general this is hard to control

We can transform any system into an LTI system through **linearization**:

↪ choose an _____

↪ apply _____

$$\dot{x}(t) = Ax(t) + Bu(t)$$
$$y(t) = Cx(t) + Du(t)$$

→ Instead of complex equations, we can now characterize the system's dynamics through _____ and the initial condition _____.

2 | Time response of a scalar LTI system

Problem statement — fully describe our system:

GIVEN $A, B, C, D,\ x(0) = x_0,\ u(t)$ FIND $x(t), y(t)$

Simplest setup: first-order, single-input, single-output $\Rightarrow$ A, B, C, D become scalar: .

$$\dot{x}(t) = ax(t) + bu(t)$$
$$y(t) = cx(t) + du(t)$$
$$x(0) = x_0$$

What do we know?

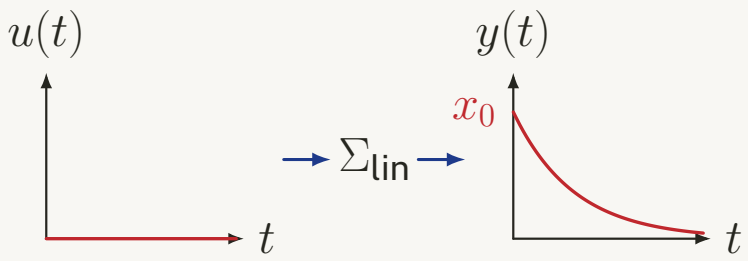
— Causality $\Rightarrow$

— Time-invariance $\Rightarrow$

We'll describe $y(t)$ using two terms:

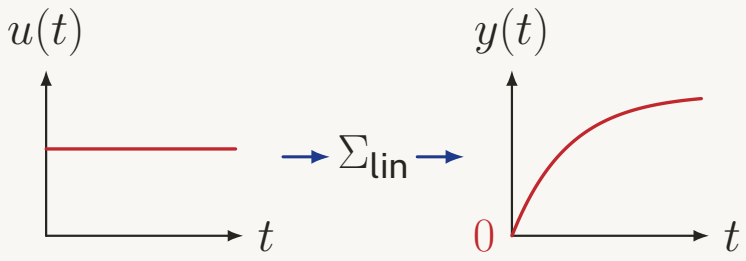
i) Initial condition response

$$\dot{x}(t) = ax(t) + bu(t)$$
$$y(t) = cx(t) + du(t)$$
$$x(0) = x_0$$



ii) Forced response

$$\dot{x}(t) = ax(t) + bu(t)$$
$$y(t) = cx(t) + du(t)$$
$$x(0) = x_0$$



⇒ Express y through y_{ic} & y_F:

i) Initial condition response

SETUP $\dot{x}(t) = ax(t)$ (I) $y(t) = cx(t)$ (II) $x(0) = x_0$ (III)

Insert into (II):

ii) Forced response

SETUP $\dot{x}(t) = ax(t) + bu(t)$ (IV) $y(t) = cx(t) + du(t)$ (V) $x(0) = 0$ (VI)

Trick: 1.Multiply both sides with e^{-at} 2.Integrate

$$\dot{x}(t) = ax(t) + bu(t)$$

Insert into (V):

Due to linearity we now have the solution to any scalar LTI system:

- $x(t) = x_{ic}(t) + x_F(t)$

$$x(t) = e^{at}x_0 + b\int_0^t e^{a(t-\tau)}u(\tau)\,d\tau$$

- $y(t) = y_{ic}(t) + y_F(t)$

$$y(t) = ce^{at}x_0 + cb\int_0^t e^{a(t-\tau)}u(\tau)\,d\tau + du(t)$$

CHEAT SHEET

Time Response

Initial condition response $(u(t) = 0,\ x(0) \neq 0)$

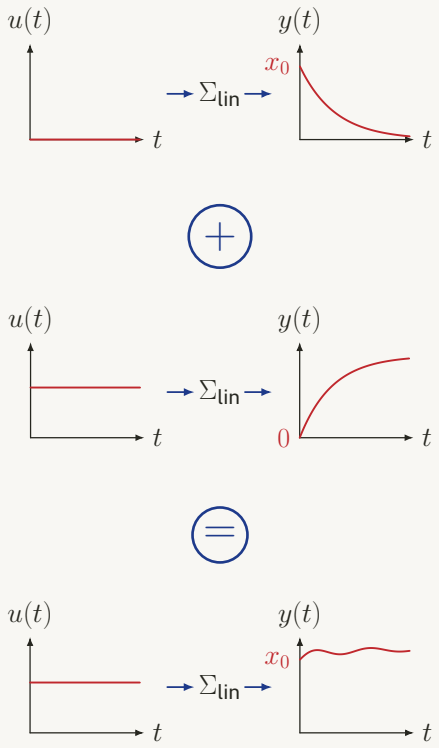
$$x_{IC}(t) = e^{At}x(0), \qquad y_{IC}(t) = Ce^{At}x(0)$$

Forced response $(u(t) \neq 0,\ x(0) = 0)$

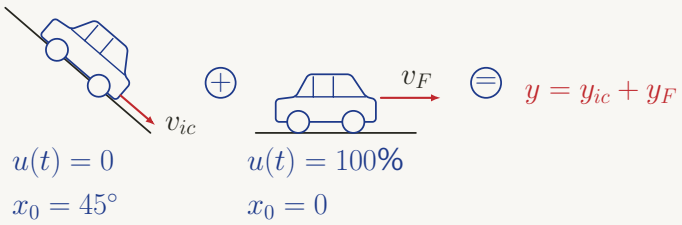
$$x_F(t) = \int_0^t e^{A(t-\tau)}Bu(\tau)\,d\tau,$$

$$y_F(t) = \int_0^t Ce^{A(t-\tau)}Bu(\tau)\,d\tau + Du(t)$$

Graphical interpretation



INTUITION



A car coasting down a slope (initial-condition response) plus a car accelerating on flat ground (forced response) — add them, and the car goes crazy fast.

Numerical example

GIVEN $a = -1,\ b = 1,\ c = 1,\ d = 0,\ x_0 = -1$

$$u(t) = h(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases}$$ (Heaviside fct.)

EX04 · TIME RESPONSE

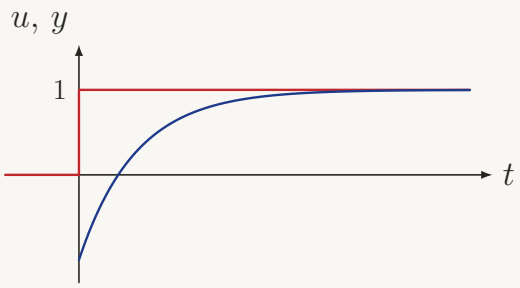
SETUP

$$\left.\begin{aligned} \dot{x}(t) &= ax(t) + bu(t) \\ y(t) &= cx(t) + du(t) \\ x(0) &= x_0 \end{aligned}\right\} \quad \Rightarrow$$

Formula for output: $y(t) = ce^{at}x_0 + cb\int_0^t e^{a(t-\tau)}u(\tau)\,d\tau + du(t)$

$\Rightarrow$ Plug in:

Graphically



... step response

3 | Generalization to a higher-order system

System matrices are, in general, non-scalar!

$$y(t) = ce^{at}x_0 + cb\int_0^t e^{a(t-\tau)}u(\tau)\,d\tau + du(t)$$

$$\downarrow$$

$\rightarrow$ How do we compute _____ ?

Intuition: 3Blue1Brown, “matrix exponentials”.

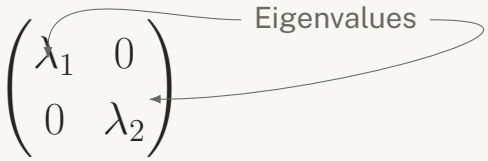
We compute them through the Taylor series expansion:

$$T(e^x) = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2} + \dots$$

For $x = At$:

$\rightarrow$ possible, but the computational effort might get out of hand.

ALTERNATIVE if A is a diagonal matrix, $A =$



then:

What if our matrix is not diagonal? Many matrices are _____.

DEF A is diagonalizable if there exists a transformation matrix T s.t. $D = T^{-1}AT$,

– $D =$ _____ _____

– $T =$ _____ _____

$\Rightarrow$ requires all eigenvectors to be _____

How can we incorporate this into our system? Our system:

$$\dot{x}(t) = Ax(t) + Bu(t)$$
$$y(t) = Cx(t) + Du(t)$$

and we found T s.t. $\tilde{A} = T^{-1}AT$. What now?

1. _____

2. Apply to system: _____

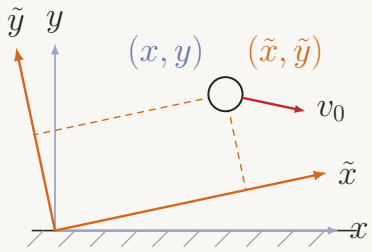
3. Simplify:

_____

4. Rename:

_____

Haven't we changed our system by changing A? — **No!** Infinitely many combinations of A, B, C, D describe the same system, transformable into each other by coordinate changes (e.g. Cartesian $\leftrightarrow$ rotated frame).



$$\begin{pmatrix} x \\ y \end{pmatrix} = T \begin{pmatrix} \tilde{x} \\ \tilde{y} \end{pmatrix}, \; T = \text{rotation}$$

Cartesian $\leftrightarrow$ rotated frame (linear, like $x = T\tilde{x}$)

Same ball, same motion v_0 — only the numbers that describe its position change.

Given

$$\dot{\tilde{x}} = \tilde{A}\tilde{x} + \tilde{B}u$$
$$y = \tilde{C}\tilde{x} + \tilde{D}u$$

we can now easily evaluate:

$$y(t) = \tilde{C}e^{\tilde{A}t}x_0 + \tilde{C}\int_0^t e^{\tilde{A}(t-\tau)}\tilde{B}u(\tau)\,d\tau + \tilde{D}u(t)$$

What if A is neither diagonal nor diagonalizable? $\Rightarrow$ bring it into **Jordan form** J (“almost diagonal”, Jordan blocks per eigenvalue).

$$J = \begin{pmatrix} J_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & J_i \end{pmatrix} \text{ where } J_i = \begin{pmatrix} \lambda & 1 & 0 & \cdots & 0 \\ 0 & \lambda & 1 & \cdots & 0 \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & \lambda & 1 \\ 0 & \cdots & \cdots & 0 & \lambda \end{pmatrix} \text{ (Jordan block for eigenvalue } \lambda_i)$$

THM if A is in Jordan form, $A = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$, then $e^{At} = \begin{pmatrix} e^{\lambda t} & te^{\lambda t} \\ 0 & e^{\lambda t} \end{pmatrix}$.

$\Rightarrow$ Allows us to again compute $y(t)$.

Computing e^{At} through the eigenvalues/-vectors of A is called .....

Back to the initial-condition response

$$y_{ic}(t) = Ce^{At}x_0 = C\,\exp\!\left(\begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} t\right)x_0$$

$$= \begin{pmatrix} C_1 & C_2 \end{pmatrix} \begin{pmatrix} e^{\lambda_1 t} & 0 \\ 0 & e^{\lambda_2 t} \end{pmatrix} \begin{pmatrix} x_{01} \\ x_{02} \end{pmatrix}$$

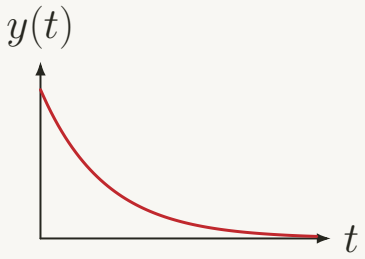
$$= C_1 e^{\lambda_1 t} x_{01} + C_2 e^{\lambda_2 t} x_{02} \quad \text{... “real” form} \rightarrow \lambda_i \in \mathbb{R}$$

Considering the more general case for complex conjugate eigenvalues, $\lambda_i = \sigma \pm i\omega$:

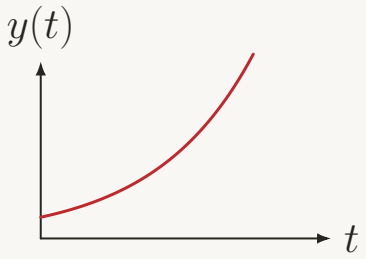
$$y_{ic}(t) = \alpha e^{\sigma t} \sin(\omega t + \varphi)$$

Depending on the eigenvalues, $y(t)$ takes the following form

i) $\lambda_i \in \mathbb{R}$**:**

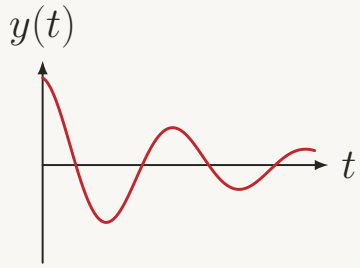


decaying — $\lambda_i < 0$

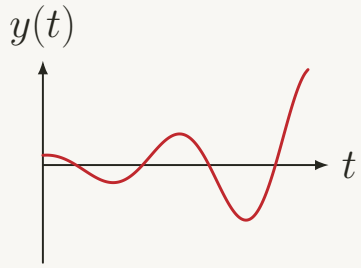


growing — $\lambda_i > 0$

ii) $\lambda_i \in \mathbb{C}$**:**



damped osc. — $\text{Re}(\lambda_i) < 0$



growing osc. — $\text{Re}(\lambda_i) > 0$

4 | Stability

We can classify three types of stability — Lyapunov, asymptotic, and BIBO.

1) Lyapunov stability

DEF a system is **Lyapunov stable** if, for any bounded initial condition & zero input, the state remains bounded:

⇒ Lyapunov stable if

2) Asymptotic stability

DEF a system is **asymptotically stable** if, for any bounded initial condition & zero input, the state converges to zero:

⇒ asymptotically stable if

3) BIBO stability

DEF a system is **Bounded-Input-Bounded-Output** stable if, for any bounded input, the output remains bounded:

... and when is a system unstable?

- A system is called **unstable** if not stable.

QUIZ

Q1) All eigenvalues of A must be > 0 for the system to be unstable.

True / False

Q2) An asymptotically stable system is always also Lyapunov stable.

True / False

Q3) A system can have one real and one complex eigenvalue.

True / False